

Solid defects and their impact on device performance

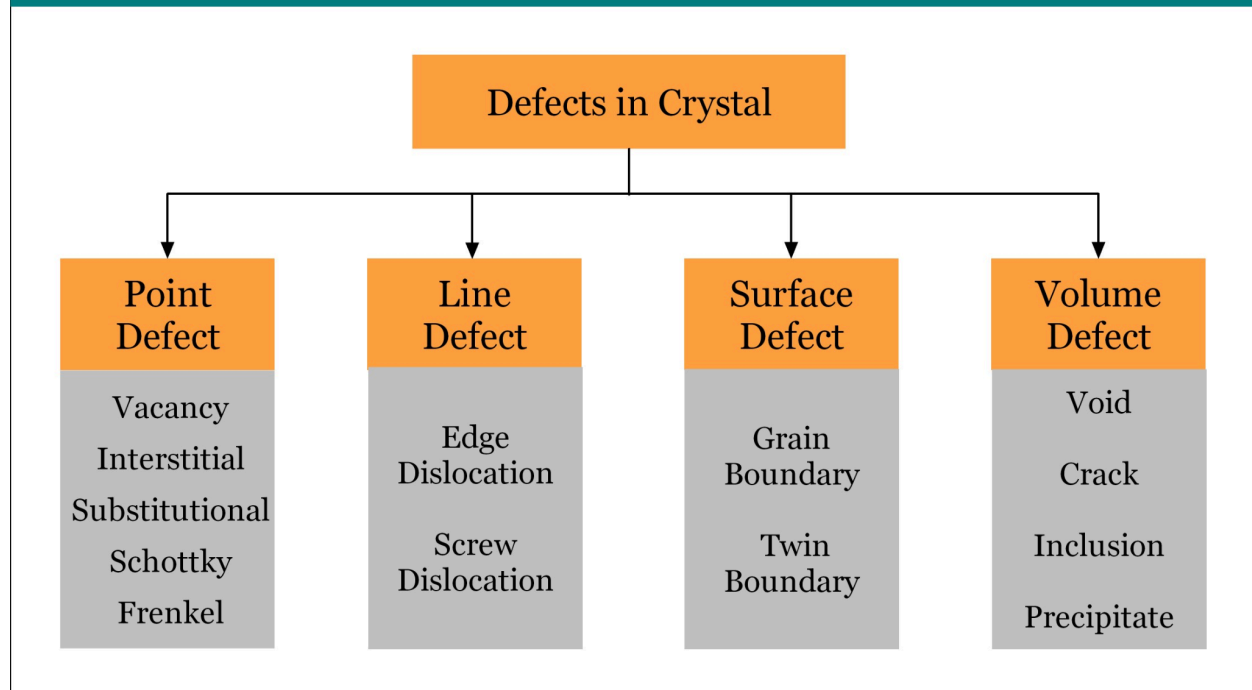
Crystals will have a regular periodic arrangement of atoms. Any deviation from this periodicity is known as defects or imperfections in crystals.

- The crystal defects affect their properties such as mechanical strength, ductility, crystal growth, dielectric strength, magnetic hysteresis, conductivity, etc.
- A perfect crystal, with every atom of the same type in the correct position, does not exist.
- Thus, all crystals have some defects. The defects in crystals may be confined to a point, line, surface and volume.

The basic classes of crystal defects are:

1. Point defects (Zero-dimensional defects)
2. Line defects (One-dimensional defects)
3. Surface defects (Two-dimensional defects)
4. Volume defects (Three-dimensional defects)

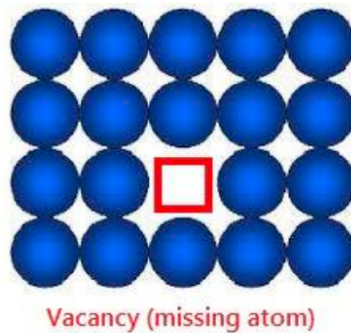
CLASSIFICATION OF DEFECTS IN CRYSTALLINE SOLID



1. Point defects (Zero-dimensional defects)

- The point defects are zero-dimensional defects
- When an atom is missing or an atom is in an irregular place in the lattice structure, the corresponding defects are known as point defects. A point defect **produces strain in a small volume of the crystal** surrounding the crystal, but does not affect the perfections of distant parts of the crystal.
- In general, point defects occur in metallic and ionic crystals. Point defects are of following types: (1) Vacancies (2) Interstitial defects (3) Substitutional defects (4) Frenkel's defect (5) Schottky's defect

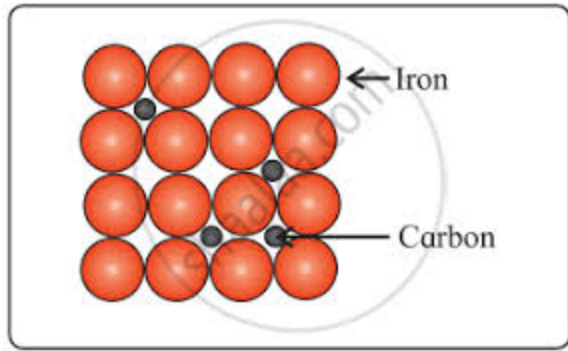
1.1 Vacancies defects: Vacancies are simplest point defects in a crystal which refers to a missing atom at its site. • Definition: The defects due to the missing atoms at their lattice sites are called vacancies. A crystal lattice with vacancy defect is shown in figure.



Vacancy defects are mainly due to the imperfect packing during the formation of crystals or due to thermal vibrations of atoms at high temperature. • At high temperature atoms frequently and regularly change their positions leaving empty lattice sites behind their positions. • For most crystals, the thermal energy is of the order of 1 eV per vacancy.

1.2 Interstitial defects

Definition: When an impurity atom tries to settle in the interstitial space between the parent atoms of the crystal without displacing, then such a defect is known as interstitial defect. • Interstitial impurities are much smaller than the atoms in the bulk matrix. • The formation of interstitial impurity defect is shown in the figure.

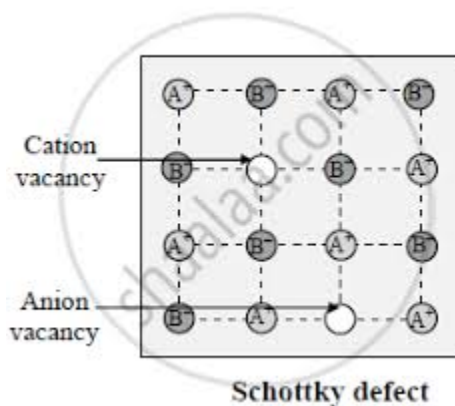


- Example: Carbon atoms are interstitial impurity atoms that are added to Iron to make Steel. In steel, Carbon atoms with radius of 0.071 nm are well fitted in the interstitial spaces between the larger Iron atoms of radius 0.124 nm.

III) In case of ionic crystals, there are two types of point defects related to vacancies. • They are: (A) Schottky defects and (B) Frenkel defects

A) Schottky defects

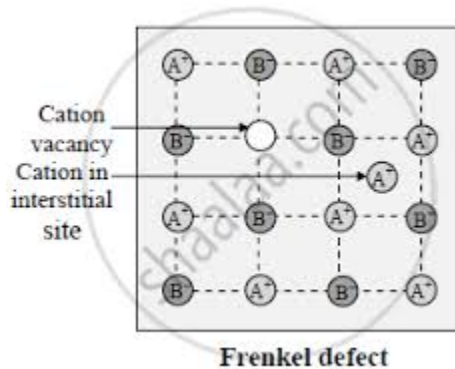
When a pair of vacancies is produced at one positive ion site and one negative ion site due to absence of positive and negative ions, then this type of defect is called the Schottky defect.



NaCl, CsCl etc., exhibits Schottky defect

B) Frenkel defects

When an ion is displaced from a regular location in the crystal lattice to an interstitial location in the crystal lattice, then this type of defect is called the Frenkel defect.



AgBr, AgCl, ZnS etc., exhibits Frankel defects.

2. Dislocations or Line defects

These are also called linear defects or dislocations. • These are one-dimensional defects

- Definition: “Dislocations are areas where the atoms are out of position in the crystal structure” (OR) “A dislocation is a one-dimensional defect around which some of the atoms are misaligned”.

- Thus, dislocations refer to a ***linear disturbance of the atomic arrangement in a crystal.***

- Dislocations are generated in crystals due to **growth accidents, thermal stress, phase transformation etc.**

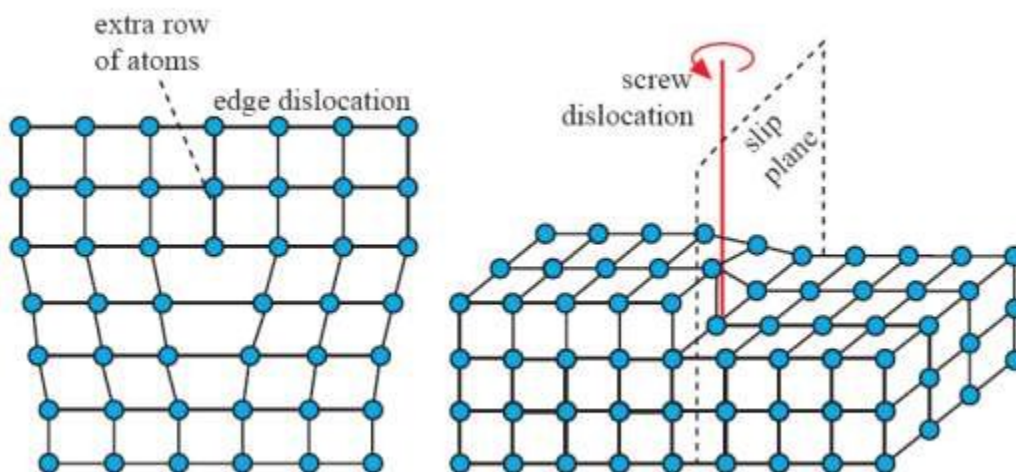
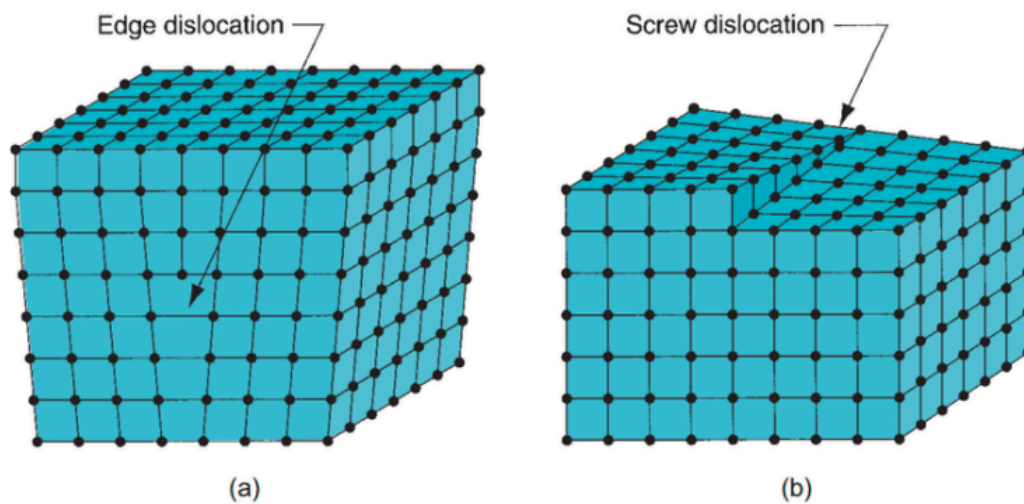
- They can be observed in crystalline materials with the help of electron microscopy.

They are two basic types of dislocations: Edge dislocations and Screw dislocations.

- **Edge dislocations:** Edge dislocation is a line defect in which an extra half-plane of atoms appears in the crystal lattice causing distortion of nearby atoms.

- **Screw dislocations**

In this case, one part of the crystal **slides over another part**. This makes the atomic layers look like a **spiral or screw shape**. Therefore it is called **screw dislocation**.



Impact of defects on material properties

Distortions of the crystal lattice often occur when impurities are added to a solid. As a result, point defects often determine the properties of a material. They can change the ease with which a material conducts electricity, its mechanical strength, its ability to be shaped by hammering (malleability), or to be drawn into wires (ductility). Dissolving small amounts of carbon in iron, for example, give the alloy known as steel, which is significantly stronger than iron. But higher percentages of carbon make steel so brittle that it can shatter when dropped.

Similarly, Dislocation defects are line defects in a crystal lattice that disturb the regular arrangement of atoms. They influence material properties by allowing plastic deformation, increasing ductility, and affecting hardness. Dislocations also scatter electrons, reducing electrical conductivity, and can create recombination centers in semiconductors. Therefore, they strongly affect both **mechanical and electrical properties of materials**.

Impact of point defect vs dislocation defect on Device Performance:

Feature	Point Defects	Dislocation Defects
Definition	Defect at a single atomic position in the crystal	Defect along a line of atoms in the crystal
Examples	Vacancy, interstitial atom, impurity atom	Edge dislocation, screw dislocation

Effect on Crystal	Disturbs a small localized region	Disturbs a long line of atoms in the lattice
Carrier Mobility	Causes carrier trapping and scattering	Causes strong scattering of carriers along the defect line
Recombination	Acts as recombination centers for electrons and holes	Strong recombination centers, especially harmful in optoelectronic devices
Leakage Current	Can increase leakage current	Can create larger leakage paths
Device Impact	Slight change in electrical properties	More severe degradation in device performance
Example Devices Affected	MOSFET, Solar Cells	Light Emitting Diodes, Solar Cells